ADVANCED MICRO DEVICES AMD

November 23, 2021 - 10:08pm EST
by kerrcap

			2021	2022
Price:	149.92	EPS	0	0
Shares Out. (in M):	1,200	P/E	55	45
Market Cap (in $M):	190,000	P/FCF	0	0
Net Debt (in $M):	0	EBIT	0	0
TEV (in $M):	187,000	TEV/EBIT	0	0

Description

AMD is a global semiconductor company selling CPUs, GPUs and other compute devices. AMD and Intel have a duopoly in the x86 CPU market, with Intel historically the dominant vendor by a wide margin. While investing in AMD after the recent run-up appears precarious, it is important to consider the business's growth drivers and its 2024 earnings power of $8-9.00+. At 25x P/E, consistent with XLNX's standalone valuation, shares should be worth $200-225, making AMD a double-digit IRR candidate from current levels.

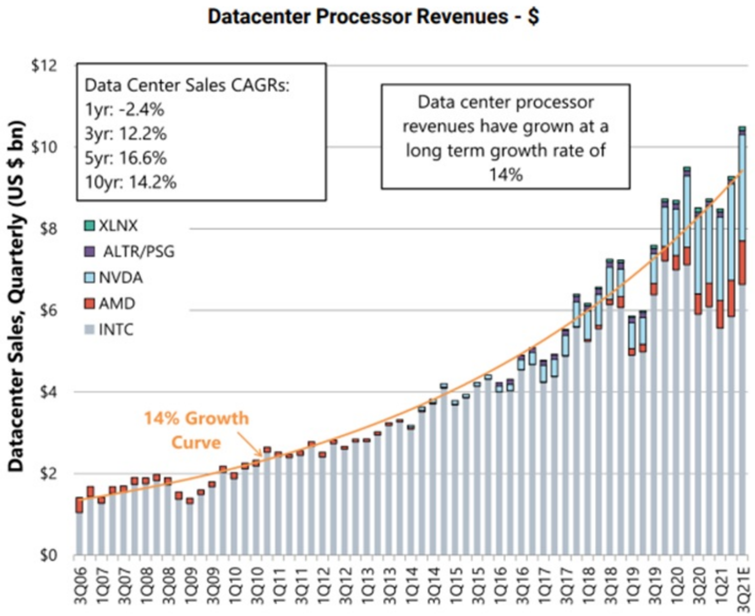
AMD's recent resurgence could be one of the greatest corporate turnaround stories in the modern era. While AMD is framed as the David in a David and Goliath story vs. Intel, we believe AMD has some unique advantages: best-in-class GPU and CPU architecture, a low-cost structure due to its early move to chiplets, leadership in core counts (96/128 core vs. 56), a TCO advantage due to superior power consumption, and a leading coherent interconnect fabric performance. With these advantages, AMD can double its market-share in the CPU Server market over the next 12 – 18 months. Server CPUs are one of the most profitable markets in semis with a 10-year growth CAGR of 14% over the past 10 years. Over the next two years, AMD is essentially supply-constrained (due to Foundry capacity), even as its main competitor Intel remains at a competitive disadvantage till at least 2024. Supply constrained growth can be very powerful, as was the case for Tesla with its manufacturing ramp or NFLX/AAPL with their gradual geographical expansion.

AMD products are now considered halo products, a huge shift in market sentiment for the business. The trajectory of core count growth from 64 cores to 96 to 128 over the next two years should drive an annual ASP uplift of ~15-20%, along with a best in industry cost structure due to its unique chiplet architecture and TSMC process node advantage. Beyond 2023, AMD should continue to compound revenue growth at a ~15% CAGR, with gross margins expanding from 50% to 62-65% and FCF growth of 30-40%.

While we remain watchful of price competition, Intel's lack of unit cost advantage and the need to generate gross profit dollars to fund the enormous $25 – 28bn annual capex (and growing) roadmap limit its ability to wage a price war to defend market share. For Intel's entire history, Intel was 12-24 months ahead in moving to a smaller transistor geometry, driving a consistent price/performance advantage. AMD historically relied on architectural innovation to drive performance gains and differentiation, while lagging in pure transistor level technical performance and unit costs due to Intel's Moore's Law leadership. In the past, Intel used this leadership in the Opteron era to price processor ASPs where AMD could simply not make an economic profit, rendering AMD's architectural advantage moot. AMD was thus relegated to niche low-end markets, with Intel dominating with a >90% market-share. This structure ended with Intel's delays at 10nm. Since both Intel and AMD are based on x86 architecture, there are minimal software compatibility issues with the introduction of AMD processors in consumer and server markets. Indeed, Intel has lost the "exorbitant privilege" of having most of the world's codebase optimized for Intel architectures.

Capitalization		Trading Multiples			
Share Price as of 11/22/21	**155.41**	**Trading Multiples**	**2021E**	**2022E**	**2023E**
Diluted Shares	1,208	EV / Revenue	11.5x	9.7x	8.5x
Market Capitalization	**187,781**	EV / Adj. EBITDA	43.7x	37.1x	33.6x
Plus: Total debt	702	EV / Adj. EBIT	48.3x	39.3x	33.5x
Less: Cash and equivalents	(3,608)	P / E	58.8x	46.8x	38.4x
Enterprise Value	**184,875**				

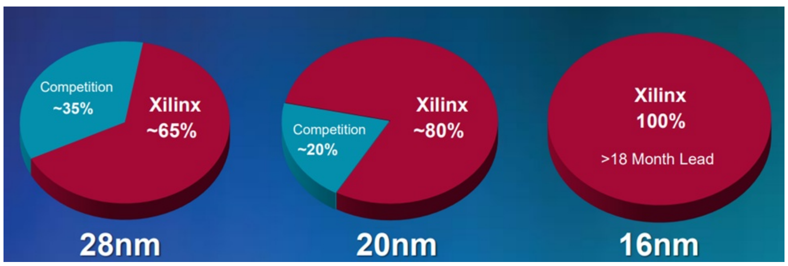
We are in a golden era of high-performance and AI compute, as evidenced by AMD Server product cadence, Intel's Alder Lake improving unit performance by 20%, NVDA iterating at a rapid rate in GPUs, AAPL demonstrating massive performance leaps with M1 silicon, and QCOM declaring its CPU ambitions recently. Like the Megahertz race a couple of decades ago, rapid performance and TCO improvements will likely drive a rapid adoption curve across most compute markets. The recent trajectory of Logic/Foundry Capex plans by TSMC, Samsung, and Intel (irrespective of national subsidies), and the capex build-outs by the cloud titans seem to confirm our hypothesis. Data Center processor revenue growth accelerated to an annual CAGR of 17% over the past five years, a trend that should persist over the next few years due to demand for AI and new compute workloads.



Source: Jefferies

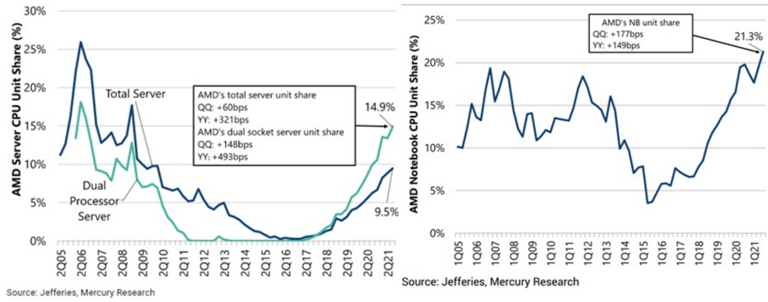
AMD's recent success against NVDA in High-Performance Compute also highlights the upside from any incremental share gains in AI compute. With AMD launching three server products over the next 18 months (Milan-X, Genoa and Bergamo), its pace of innovation and improvement is unprecedented. Intel's Sapphire Rapids Server CPU is a solid incremental update, but AMD will maintain leadership in core counts (96/128 core vs. 56) and a 50-60% TCO advantage due to superior power consumption. The extent of AMD's upcoming dominance in Servers continues to be underappreciated. We note recent news of Meta deploying AMD Server CPUs (Genoa), Microsoft deploying the new Milan-X Servers in Azure Cloud, and Google deploying Milan Server chips for scale-out Server workloads. Meta's decision to move to AMD is a tectonic shift as Meta was an entrenched Intel customer for many years. The earliest Intel can respond to AMD competitively is 2024 due to publicly provided foundry roadmaps. We believe a 35-40% market-share for AMD in Servers should be considered an absolute floor as we consider the company outlook over the next few years.

We believe some investors are still anchored to AMD's prior peak server market-share of 25%, where Intel successfully leveraged its Moore's Law driven unit-cost transistor leadership to price AMD out of many markets. But today's environment could cause a disruptive dislocation to the Server ecosystem, with a non-linear impact to market-share. We see similarities to Xilinx with its fabless leading-edge technology development, leveraging TSMC as a foundry, a duopoly-type industry structure in FPGAs, followed by Intel's missteps in technology execution that resulted in Xilinx expanding its market-share from low-40% to over two-thirds of the overall FPGA market. Every subsequent technology transition drove higher incremental market-share for Xilinx. Similar to Xilinx, a time to market advantage at leading-edge technology nodes and architectural advantages should drive majority of the share gains for AMD going forward.



Source: Xilinx

In Servers, AMD remains in the sweet spot of ramping Milan-X next year, followed by Genoa in 2H22 and Bergamo in 1H23. According to Mercury Research, AMD's total server unit market share increased to 9.5% in 3Q21 (+3 pts Y/Y), and we expect this accelerating trend in share-gains to persist through 2023. In PCs, AMD remains supply constrained, and is targeting high-end commercial workstations through 2022. With Notebook unit share exceeding 20% in 3Q21, AMD is still largely targeting the high-end PC market due to ongoing supply-constraints likely limiting the extent of market-share gains in this market.



Source: Mercury Research, Jefferies

AMD's chiplet technology is disrupting the Server and general compute markets. Historically, server chips were manufactured as a massive, single piece of silicon that lowered yields and increased unit costs. AMD's chiplet approach separates the server chip into multiple smaller dies with lower unit costs and higher yields. This cost analysis from a previous report compares AMD Rome Server chip's cost structure to Intel's Skylake product line. These cost differentials are amplified at higher core counts. According to the analysis below, AMD's Epyc Rome server chip had a 30-35% unit cost advantage vs. Intel Skylake.

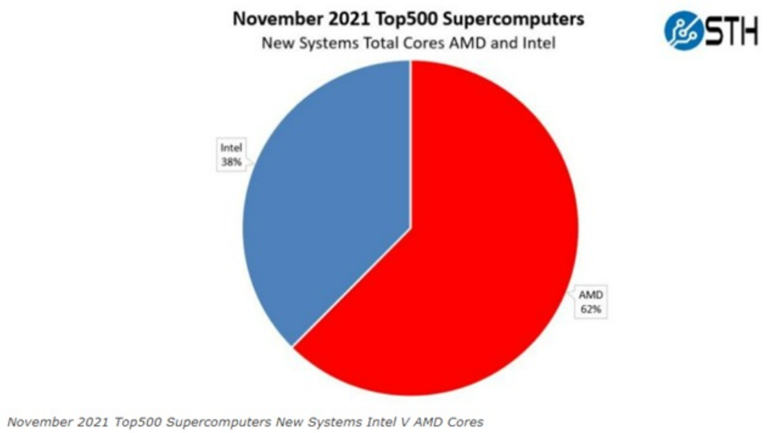
Server Chip Cost Analysis	Intel	AMD
Part	Skylake SP	Epyc Rome
Die Size	698mm^2	70mm^2
Est DPW	76	862
Est Good Die/Wf	26.6	775.8
Est.Manufacturing Cost/Wafer	$3,500	$4,000
# Die/Server	1	8
# cores/Server	28	64
Est Core Cost/Server	$132	$41
# I/O Die	0	1
Est I/O Cost/Server	$0	$27
Est Total Server Chip Cost	$132	$68
Est. Packaging Cost	$30	$40
Est Total Server CPU Cost	**$162**	**$108**

Source: Eagle Point Global

In GPUs for data centers, AMD's ability to combine both a leading-edge CPU and a leading-edge GPU remains underappreciated. AMD is looking to leverage its multi-die capability against a monolithic die competitor, doing in GPUs to NVDA what it previously did in CPUs to Intel. The Genoa server ramping in 2H22 will likely support a new high-speed Infinity fabric with better support for heterogeneous accelerators including GPUs, making AMD GPUs a first-class citizen in AMD's compute systems.

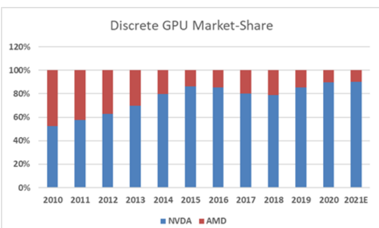
Recent shifts in the high-performance computing GPU world are quite informative. AMD achieved a majority incremental share in HPC deployments, with AMD deploying both CPUs and GPUs. HPC market is potentially a leading indicator of mainstream adoption, with Nvidia's success in HPC GPU Compute deployments a precursor to its recent success in AI compute. AMD won multiple major supercomputer wins in the United States, including some of the biggest deployments like El Capitan and Frontier. In just over a year, AMD tripled its supercomputer deployments from 21 to 73, now driving almost two-thirds of new system cores in HPC deployments. AMD's recent CDNA 2 GPU release trounces NVDA's leading Ampere GPU in HPC benchmarks. While small in absolute dollar contribution, we note the recent inflection in AMD's Datacenter GPU revenue.

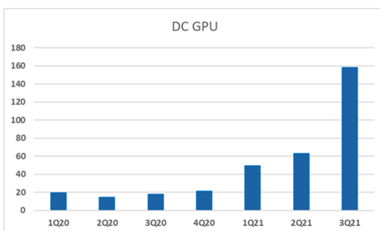
We are not claiming that AMD's ROCm software stack is in any position to take market-share against CUDA anytime soon in AI benchmarks. But with the support of mixed-precision AI workloads in the new CDNA2 GPUs, many cloud providers will take the compute performance offered, while leveraging internal compiler efforts to run AI frameworks like Tensorflow or Pytorch. The launch of Genoa Server chip in 2H22 will likely be a key catalyst to watch.



Source: ServeTheHome

In Gaming GPUs, the opportunity set remains attractive. AMD's discrete Gaming GPU market-share is below 10% currently due to supply constraints and historical missteps in the RX Vega GPU family. AMD has a competitive roadmap and could easily double or triple its market-share should supply normalize in discrete GPUs. However, due to priorities in supply allocation (servers are produced first, followed by consoles and then commercial workstations), we believe AMD will not target the mainstream GPU market till at least 2H23.





Source: Mercury Research, AMD

Valuation

The majority of AMD's valuation is dependent on the ramp in Servers. Assuming a 45% market-share in a couple of years, without any aggressive share gain assumptions elsewhere, we see AMD reaching a run-rate EPS potential of $8-9/share. The analogy to Xilinx likely holds in valuation as well. Xilinx was consistently valued at around 25x EPS/FCF multiple as an independent company, and we use that as a framework in valuing AMD. Thus, we see upside to AMD shares to $200-225, without any aggressive upside from GPU share gains.

Revenue ($M)	2018	2019	2020	2021	2022	2023	2024
Desktop CPU	1,098	1,787	1,949	2,209	2,231	2,219	2,207
Notebook CPU	861	1,318	2,839	4,274	4,723	5,266	5,661
Total CPU	1,960	3,105	4,788	6,483	6,954	7,484	7,868
GPU	2,165	1,504	1,524	2,647	3,406	4,028	4,764
Licensing		100	120	120	207	300	435
Total C&G Revenue	**4,125**	**4,709**	**6,432**	**9,250**	**10,567**	**11,813**	**13,066**
Semi-Custom	1,550	779	1,347	2,939	3,293	2,717	2,241
Server	352	849	1,549	3,282	6,260	10,742	15,361
Embedded	448	394	435	737	800	800	800
Total EESC Revenue	**2,350**	**2,022**	**3,331**	**6,958**	**10,353**	**14,258**	**18,402**
Total Revenue	**6,475**	**6,731**	**9,763**	**16,209**	**20,920**	**26,071**	**31,469**

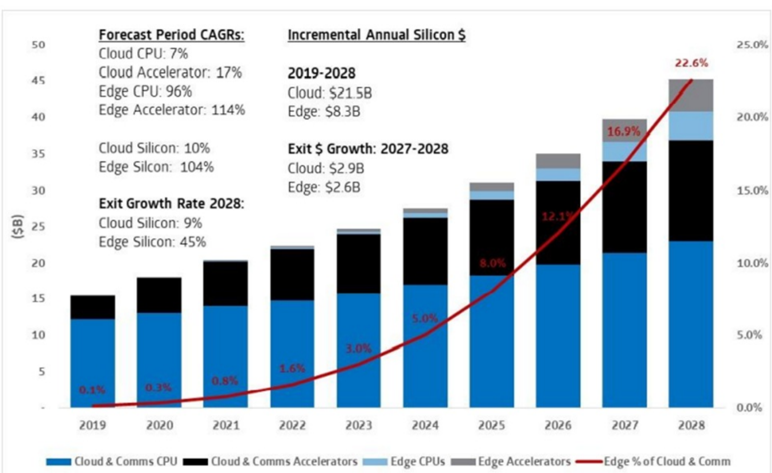
Revenue ($M)	2018	2019	2020	2021	2022	2023	2024
Total Gross Margin (%)	**37.1%**	**42.6%**	**44.5%**	**47.6%**	**52.8%**	**57.9%**	**61.7%**
Incr. Gross Margin (%)		*181.9%*	*48.9%*	*52.4%*	*70.6%*	*78.8%*	*79.5%*
Gross Profit ($)	2,400	2,865	4,347	7,723	11,047	15,108	19,401
R&D	1,434	1,547	1,983	2,853	3,473	3,937	4,752
SG&A	562	738	981	1,410	1,694	2,007	2,423
Opex	**1,996**	**2,285**	**2,964**	**4,263**	**5,167**	**5,944**	**7,175**
% Rev							
R&D	*22.1%*	*23.0%*	*20.3%*	17.6%	16.6%	15.1%	15.1%
SG&A	*8.7%*	*11.0%*	*10.0%*	8.7%	8.1%	7.7%	7.7%
Opex	***30.8%***	***33.9%***	***30.4%***	***26.3%***	***24.7%***	***22.8%***	***22.8%***
EBIT	404	580	1,383	3,460	5,879	9,164	12,226
EBIT (%)	***6.2%***	***8.6%***	***14.2%***	***21.3%***	***28.1%***	***35.1%***	***38.9%***
Incr. EBIT %		*69.1%*	*26.5%*	*32.2%*	*51.4%*	*63.8%*	*56.7%*
Interest Expense	121	199	108	70	60	40	40
PBT	283	381	1,275	3,390	5,819	9,124	12,186
Taxes	-9	31	43	339	698	1,095	1,462
Tax Rate	*-3.2%*	*8.1%*	*3.4%*	10.0%	12.0%	12.0%	12.0%
Net Income	292	350	1,232	3,051	5,121	8,029	10,724
EPS ($)	**$ 0.25**	**$ 0.29**	**$ 1.00**	**$ 2.47**	**$ 4.11**	**$ 6.42**	**$ 8.57**

Sources: AMD filings, our analysis

Risks

The PC Cycle. While PCs remain a key debate, our sense is AMD will be supply constrained and a share gainer through 2023. A number of industry analysts are calling for a higher refresh rate driving sustained strength as Enterprise recovery / hybrid work / Leasing / Windows refresh / Gaming notebook growth creates a stable demand in the PC space. For context, a reduction in the PC installed base refresh rate by 6 months (from 4.5 years to 4 years) drives a 10%+ upside to annual PC unit shipments.

Accelerator competition. The CPU will likely remain the main compute tool even as we enter the era of domain specific accelerators and GPU compute, as evidenced by this forecast by Cowen. We thus see overall CPU compute demand continuing to grow at a double-digit CAGR, with AMD's share gains in GPU compute representing a source of further upside.

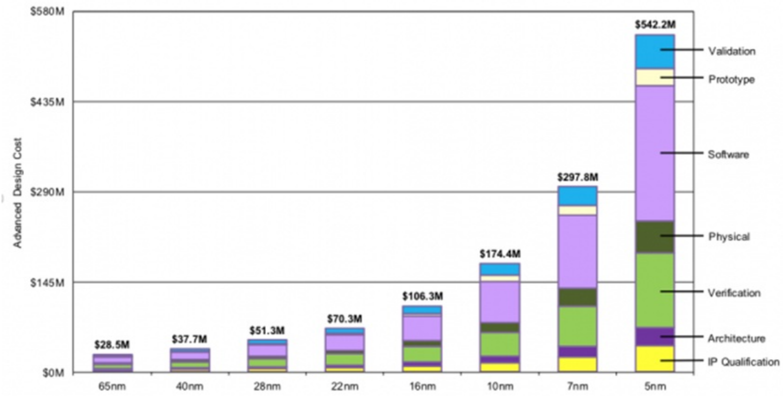


Source: Cowen and Company estimates, Tolaga Research

Cloud Chips. The decision to design chips internally or source via merchant silicon is an implementation question. The cloud vendors still rely on ASIC companies to provide the full chip capability. As demonstrated by AVGO and MRVL's margin outlook, a cloud-centric custom ASIC business does not have inferior unit economics to the merchant silicon model. Due to its Gaming console heritage, AMD brings extensive experience and deep relationships designing custom ASICs. With chip development costs now running over $500M for a single chip design vs. $50M over a decade ago, the scale advantage continues to tilt towards an at-scale merchant business.

ARM. ARM needs a material cost or performance advantage to win over x86. With AMD's new Server chips offering core counts comparable to leading ARM vendors, the unit cost advantage currently offered by high-core count ARM servers should dissipate. We believe x86 continues to lead in single-threaded performance and the existing x86 code base remains a considerable incumbent advantage. That said, AMD is an ARM licensee with previous experience designing both custom and standard ARM cores.

AMD/XLNX Combo: We think concerns around a "mature FPGA" market slowing down overall long-term AMD/XLNX growth rates below 20% are misplaced. As the figure below demonstrates, the cost to develop a digital chip has grown from $50M a decade ago to over $500M with the latest 5nm technology node at TSMC. Due to rising design complexity and growing verification costs, we believe FPGAs outperform and displace ASICs longer-term as it is getting more expensive to design new chip ASICs. Said differently, it is easier to build general-purpose processors (GPU/CPU/FPGA) and amortize the costs over a wide variety of workloads and end-markets.



Source: IBS

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Catalyst

Incremental supply from TSMC, Genoa Server ramp in 2H22, and the 128-core Bergamo cloud-optimized Server in 1H23, all driving accelerating share gains and design win activity

Messages

Subject Xilinx
Entry 11/24/2021 10:09 AM
Member zamperini

Thank you for the writeup on AMD. Considering the current price differential between AMD and XLNX, do you think buying XLNX can provide a better entry point to owning AMD? Please share your views on the acquisition considering Lisa, per her 3Q commentary is still aiming for end of 2021 to complete the merger.

Subject Re: Xilinx
Entry 11/24/2021 01:05 PM
Member kerrcap

We aren't yet informed enough to comment on the probability of deal closure. The spread seems reflective of the many risks involved in obtaining approval along with XLNX's material downside should the deal become blocked for political concerns. With that said, China appears not to have blocked an M&A transaction where both companies were based in the U.S., and AMD and XLNX both have a strong presence in China.

Subject Supply Constraint/Market share assumption/ARM/XLNX
Entry 11/30/2021 06:46 AM
Member Hamilton1757

Thanks for the write up on AMD, I also think the chiplet design is very interesting, increasing yield and lowering cost (in a sense they can even recycle some partially defective dies, to use in lower-end product line). A few questions if I may:

- How long do you think the supply constraint will further last? Think you implied 2H23 in the article, has this changed post Omicron? Do you think the supply constraint is actually buying extra time for INTC?

- Do you think 45% server market share in a couple of years is realistic? I probably won't question the 45% number but rather the timeline. It took AMD 3 years to ramp back to 10% total server market share (Jefferies/Mercury graph), with supply constraint still in place for the next couple of years is 35% increase of market share achievable? (which is ~3x the speed of 17-20 period)

- ARM threat to x86? although on the server CPU side this may still seem distant, the progress on PC CPU side has been demonstrated by AAPL's M1 silicon. Do you have any view of ARM vs x86 going forward? What's the main advantage of x86 against ARM?

- On GPU side, AMD's software stack is quite weak, as I think AMD do not have extra resource to invest in their open software platform. It's also worth noting that NVDA do not have the advanced node constraint like INTC has (NVDA also uses TSMC) and NVDA usually uses quite mature nodes which means further performance boost is guaranteed when they switch to new node. So I think on the software side NVDA may have 5-10yrs lead due to their extensive investment and collaboration with the scientific world and on the hardware side NVDA also have defensive cards in hand. AMD's ability to combine both CPU and GPU might seem underappreciated but in HPC I think nothing is preventing AMD CPU to be combined with NVDA CPU?

- AMD/XLNX deal: what do you think the impact would be on AMD if the deal is approved by regulators? XLNX EPS (if we consider the 1.7234 exchange factor) is actually lower than AMD and the growth is by far lower, would this deal impact your PT and how much?

Subject Re: Supply Constraint/Market share assumption/ARM/XLNX
Entry 11/30/2021 03:49 PM
Member kerrcap

1. We do expect industry-wide supply of 5nm (TSMC) to remain tight through 2022, with further supply chain disruptions extending that timeline into 2023 and beyond. Our reference to H2 2023 addressed AMD’s ability to target mainstream GPUs and wasn't meant as a goalpost for global supply. And yes, we'd agree that access to substrate and IDM capacity benefits INTC as the incumbent.

2. Using AMD's 2.5pts of market share gain (q/q) in Q3, a linear extrapolation from 15% results in almost 40% by end of 2023. It did take three years to reach 10% market share, but this reflects the deployment hurdles for any vendor in the cloud. Even NVDA's AI GPU took a few years to reach critical mass.

3. From an instruction set perspective, there’s minimal differences between x86 and ARM. These differences arise in the microarchitecture and that’s more of an implementation function.The cloud vendors will continue building custom ARM servers, but AMD and INTC now offer custom CPU capability and can match the roadmap/IP of ARM in most circumstances. Apple’s ability to co-design hardware and software along with M1 silicon does provide its advantages, but we expect AMD to respond with its own competitive roadmap. And If you compare the gross margins of custom ASIC vendors like MRVL/AVGO, they're similar to merchant suppliers like AMD. We think this augurs well for AMD and fears of margin compression may be overdone.

4. The world has come full circle on CUDA. Back in 2017/2018 it seemed that AI frameworks like Tensorflow would dis-intermediate CUDA. But now you might be overstating the ability of NVDA libraries to maintain a competitive differentiation for NVDA over 5+ years. And even if we allocate two-thirds of the market to NVDA, there’s plenty of addressable market for AMD and others to grow into. Yes, NVDA offers their GPU systems using AMD processors, but we expect NVDA to move to their internal Grace Server CPUs. These are meant to complement NVDA's GPU offerings and won't compete directly with CPUs on a standalone basis. And while the market for HPC is smaller than AI, AMD has won a number of supercomputer deals in HPC, effectively shutting out NVDA.

5. At the time of the initial deal announcement, XLNX would have doubled AMD’s EBITDA. As for growth rates, XLNX likely grows in the mid-teens if they can accelerate the Datacenter business. XLNX guided to $1bn of datacenter revenue by 2023 – that might slip by a year, but it alludes to the direction of future growth. XLNX’s adaptive platform strategy (Zynq/RFSoC/ACAP) has been a powerful growth driver, with platform revenue growing as a share of overall revenue. The Vitis software platform from XLNX is much further along vs. AMD's ROCm. XLNX also has a long history in hardware/software interface compiler software development (starting with Vivado/HLS). Lastly, XLNX pioneered 3D packaging at TSMC and should help AMD drive growth in the Networking/SmartNIC verticals.